# Zurab Jashi

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## Education

**Perimeter Institute** — Waterloo, ON
*MSc in Physics, GPA: 4.0* — *Sep. 2023 – April 2025*
*Relevant coursework*: Quantum Field Theory, Quantum Information, Statistical Physics, String Theory, Mathematical Physics, Quantum Physics, Standard Model, Condensed Matter, Gravitational Physics

**New York University** — New York & Abu Dhabi
*BSc in Mathematics, GPA: 3.94* — *Aug. 2019 – May 2023*
*Relevant coursework*: Multivariable Calculus, Linear Algebra, Abstract Algebra, ODEs, PDEs, Chaos & Dynamics, Advanced Probability, Stochastic Calculus, Computational Physics, Topology, Differential Geometry, Theoretical Machine Learning, Applied Machine Learning

## Experience

**AI Engineer** — Abu Dhabi, UAE
*City Football Group (Manchester City)* — *Aug. 2026 – Present*
- Build and maintain AI agents using Pydantic AI, with a FastAPI backend, React/Next.js frontend, and PostgreSQL database, deployed on Azure.
- Develop a *"Partnership Agent"* that helps the partnerships team discover and evaluate new prospect companies.
- Build a *"Treasury Agent"* that classifies transactions and assists the finance team with reporting.
- Contribute to *"Skeleton"*, the shared framework underlying our agents, and support the rollout of two additional agents for further teams.
- Partner closely with stakeholders across the *Enterprise Solutions* team, handling reporting and resolving support tickets.

**AI Engineer** — Abu Dhabi, UAE
*New York University* — *Jan. 2026 – Aug, 2026*
- Designed and implemented a multi-agent architecture using LangGraph to discover state-of-the-art, mathematically worthwhile problems, focused primarily on probability theory.
- Worked closely with mathematicians to run experiments, fine-tune models, and optimize agent performance.
- Built a FastAPI + Next.js agent to help the mathematics team organize and manage their research projects.

**Big Data Engineer** — Tbilisi, Georgia
*Bank of Georgia* — *Apr. 2025 – Jan. 2026*
- Developed Spark pipelines (batch + streaming) to calculate customer eligibility for cashback, discounts, and loyalty offers across 4M+ customers.
- Built real-time Kafka workflows that update customer eligibility instantly when their data changes.
- Supported campaign launches by rerunning eligibility logic on demand for newly created offers.
- Handled large data volumes during events like Black Friday (up to 1.5B rows), improving reliability and performance.
- Worked with QA to investigate customer cases and fix logic issues in complex Spark code.
- Deployed updates from dev to prod using Git, Jenkins, Docker, and Kubernetes.

**Machine Learning Research Intern** — Waterloo, Canada
*Perimeter Institute for Theoretical Physics* — *May 2022 – Jan. 2023*
- Selected as one of 12 interns from 400+ applicants to work with Sebastian Wetzel on *Twinned Regression*, a novel ML method.
- Implemented and tested the algorithm, which restructures inputs from $x \to y$ to $(x_1, x_2) \to (y_1 - y_2)$, enabling ensemble-like behaviour and uncertainty estimation.
- Applied the method across neural networks, NN+kNN hybrids, and extensively on random forests.
- Ran large-scale experiments on the Graham supercomputer (Compute Canada); observed 5–10% accuracy gains on some datasets and underperformance on others due to increased input dimensionality.
- Contributed to extending the method to a semi-supervised version; results published at arxiv.org/abs/2301.01383.

## Projects

**Co-founder & Product Lead**
*Bulbee* — Continuous
- Co-founded and serve as product lead and co-developer of Bulbee, an innovative tutoring platform live at bulbee.ge connecting tutors and students across Georgia.
- Built the full Spring backend and Next.js frontend with fully integrated payment system for bookings.
- Drive product direction and growth marketing across social media platforms to grow the platform into a leading business in the education sector.
- Leading the upcoming rollout of AI-assisted learning features for students.

## Skills

**Programming Languages** - Python, Java, Typescript
**Data** - Kafka, Pydantic, HDFS, PostgreSQL
**Frameworks** - FastAPI, Next.js, Spring, Spark, Airflow
**Infra & CI/CD** - Azure, Cypress, Github Actions

## Awards

2023 - **Full Scholarship** at *MSc Program at Perimeter Institute*
2021 - **Silver Medal** at *International Mathematics Competition*
2019 - **Full Scholarship** at *BSc Program at New York University*
2019 - **Silver Medal** at *International Physics Olympiad*
2018 - **1st place** at *Georgia's National Physics Olympiad*
2017 - **Bronze Medal** at *International Physics Olympiad*